

Ag. Nikolaos WWTP Upgrade & Expansion

Electrical Design, Technical Studies & Project Supervision

Project Description

The Upgrade and Expansion of the Ag. Nikolaos Wastewater Treatment Plant (WWTP) was developed to support the modernization, operational reliability, automation, monitoring, and future expansion requirements of the wastewater treatment infrastructure. The project included MV/LV electrical power distribution systems, transformer and generator selection and installation, automation architecture, PLC communication systems, control panels, instrumentation systems, 2D and 3D engineering design, technical presentations, and supervision of electrical and electromechanical installations. Particular focus was given to electrical reliability, protection coordination, system monitoring, operational safety, commissioning preparation, and integration of automation and instrumentation systems for efficient plant operation.

My Role

Electrical Design Engineer, Project Supervisor & 3D Design Coordinator Responsible for electrical design studies, MV/LV system coordination, technical documentation, transformer and generator sizing support, 3D engineering design, technical presentations, automation coordination, and supervision of electrical installation works. The role also included coordination with multidisciplinary engineering teams, support during implementation phases, and technical supervision of electrical infrastructure and automation systems.

Scope of Work

- Electrical studies and technical documentation
- MV/LV electrical systems coordination
- Transformer and generator selection and installation
- 2D and 3D engineering design
- Technical presentations and project visualization
- Automation and PLC coordination
- Instrumentation systems integration
- Electrical layouts and engineering documentation
- Site supervision and engineering support
- Testing and commissioning support

Software & Tools

- 2D Autodesk Design
- 3D Autodesk Design
- EPLAN Electric P8
- Microsoft Office
- Revit
- TiSoft
- 4M Adapt
- TIA Portal
- DIALux
- EcoStruxure Power Design

Selected engineering drawings, multiline diagrams, technical studies and documentation included in this portfolio are presented for demonstration purposes only. Sensitive contractor information, signatures, and company details have been removed.